

SESSION 5

CQs:

1. CQ33: Which road signs in the dataset have a circular shape and a red border?
2. CQ34: Which individual signs are classified by the reasoner as Danger Warning Signs?
3. CQ35: Which road signs possess a blue ground colour?

LLM: Gemini

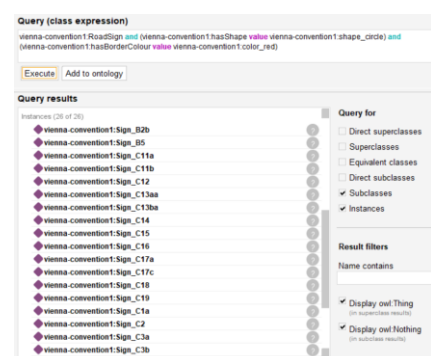
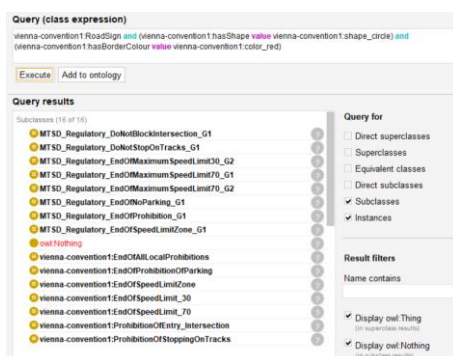
LLM Prompt: Role: You are an expert Ontology Engineer. Objective: Formalize natural language Competency Questions (CQs) into Description Logic (DL) Queries using Manchester Syntax, so they can be executed in the Protégé DL Query tab. Scope: The road sign ontology based on the Vienna Convention (Namespace: vienna-convention1:). Tasks: Translate the following CQs into DL Queries:

1. Which road signs in the dataset have a circular shape and a red border?
2. Which individual signs are classified as Danger Warning Signs?
3. Which road signs possess a blue ground colour? Output Format: Provide the formal DL query strings corresponding to each question.

Formalisation & Answers:

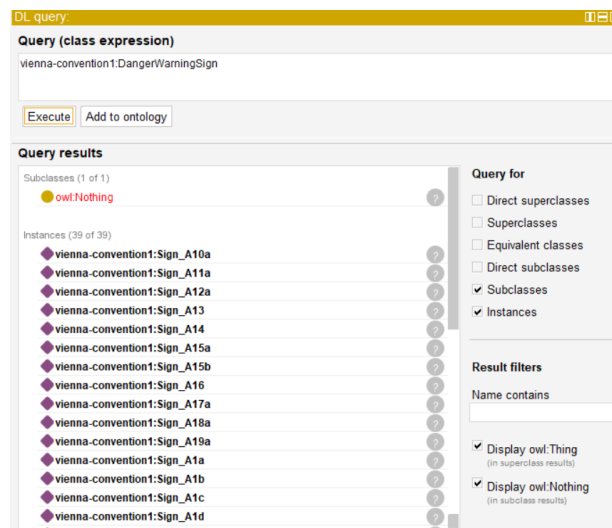
CQ33: Which road signs in the dataset have a circular shape and a red border?

- **DL Query:** `vienna-convention1:RoadSign and (vienna-convention1:hasShape value vienna-convention1:shape_circle) and (vienna-convention1:hasBorderColour value vienna-convention1:color_red)`
- **Answers (Reasoner Output):** The reasoner successfully returned 16 subclasses and 26 instances matching these exact conditions (representing the Prohibitory signs dataset).



CQ34: Which individual signs are classified by the reasoner as Danger Warning Signs?

- **DL Query:** vienna-convention1: DangerWarningSign
- **Answers (Reasoner Output):** The reasoner inferred and returned 39 instances belonging to this class based on their underlying shape and colour properties.



CQ35: Which road signs possess a blue ground colour?

- **DL Query:** vienna-convention1: RoadSign and (vienna-convention1: hasGroundColour value vienna-convention1: color_blue)
- **Answers (Reasoner Output):** The query returned 26 subclasses and 16 instances (representing Mandatory and Information signs sharing this ground colour).

